

FEATURE ENGINEERING

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**01 BUILD YOUR OWN
RESEARCHER**

02 YOUR STRATEGY

**03 CURATOR, LIVE —
THEN STOP**

**04 WHAT TO LOOK FOR
IN THE EXAMPLE**

03

A researcher of your own, a strategy of your own, and the data it needs — on disk before you leave.

WHO READ THEM.

Hands up, work by work. Then one line from whoever read it: what did it change about your objective?

TIMELINE 01 · THE ARGUMENT

- ☐ 1970 Fama · Efficient Capital Markets: A Review of Theory and Empirical Work
- ☐ 1979 Kahneman & Tversky · Prospect Theory
- ☐ 1980 Grossman & Stiglitz · On the Impossibility of Informationally Efficient Markets
- ☐ 1997 Shleifer & Vishny · The Limits of Arbitrage
- ☐ 2004 Lo · The Adaptive Markets Hypothesis

TIMELINE 02 · THE EVIDENCE

- ☐ 2011 Khandani & Lo · What Happened to the Quants in August 2007?
- ☐ 2016 Daniel & Moskowitz · Momentum Crashes
- ☐ 2016 McLean & Pontiff · Does Academic Research Destroy Stock Return Predictability?
- ☐ 2016 Harvey, Liu & Zhu · ... and the Cross-Section of Expected Returns
- ☐ 2023 Jensen, Kelly & Pedersen · Is There a Replication Crisis in Finance?

THE RULE FROM SESSION 02 A paper you did not read is a claim you cannot cite. Mark the board; the unread ones go into your researcher this week.

WHAT DID IT CHANGE.

Three questions, answered out loud. Write your answer down — it becomes the first claim in your `OBJECTIVE.md` this afternoon.

01

WHICH CLAIM CHANGED YOUR MIND?

One paper, one claim, and the belief you held before it. If nothing moved, say which paper you would argue with.

02

WHICH ONE ARGUES AGAINST YOUR OBJECTIVE?

Crowding, crashes, decay, failed replication — four of the ten exist to attack the idea you brought. Which one lands?

03

WHAT WOULD YOU TEST FIRST?

The smallest experiment that could kill your idea. That is the signal you are looking for in the example this week.

THIS SESSION

INSTALL, NAME, SCAFFOLD, **DOWNLOAD.**

Hands-on, start to finish: a researcher of your own, a strategy of your own, and its data on disk — then we stop, on purpose, at the security master.

BUILD YOUR OWN RESEARCHER

Paste one line → name it → the interview → feed it the ten works from session 02

ONE PER PERSON, NOT PER STRATEGY.

Your reading is yours and outlives any one idea. Each strategy is a separate repository from the process template.

YOURS, ONE OF THEM

THE RESEARCHER

Sources/ · Knowledge/ · Philosophy/ · Projects/

Reads every strategy's Bibliotheca/, drafts its claims and hypotheses, and never writes into one uninvited.

STRATEGY A

MOMENTUM IN MID CAPS

Its own repo, its own universe, its own bibliography.

STRATEGY B

QUALITY AFTER EARNINGS

Same eight steps, same folder names, different idea.

STRATEGY C

THE ONE YOU WROTE LAST WEEK

Your objective from session 02 lives here.

TWO REPOS, TWO JOBS

your researcher = what you know · your strategy = what you are testing · one package builds both

FOUR FOLDERS, ONE DIRECTION.

The arrow only points one way. That is what stops the library becoming a second opinion of itself.

Sources/

WHAT YOU READ

Books, Papers, Clippings. The researcher reads here and never writes here. PDFs stay out of git.

Knowledge/

WHAT IT READ

One note per chapter read, and a wiki page per idea. INDEX.md is read first, always; LOG.md is append-only.

Projects/

WHAT YOU ASKED FOR

Source notes, objective and blueprint drafts, lessons. You copy them into the strategy yourself.

Philosophy/

YOUR VOICE — THE SIDE CHANNEL

How you invest, in your own words. Cited by everything, generated from by nothing, and edited only through a diff you approve. Anything you would not want read does not go in the repository at all.

PASTE ONE LINE.

It reads the repository's setup notes and does the rest. It asks you for one thing: the name — and the name is the folder.

PASTE THIS INTO CLAUDE OR CODEX

Please help me install this repo: <https://github.com/KaxaNuk/KaxaNuk-Researcher>

Behind the line: `uv tool install apm-cli`, then `apm install -g KaxaNuk/KaxaNuk-Researcher --target claude`. One package, installed once for your user — it carries the researcher, every Lab skill, the strategy template and the worked example. Every folder you open afterwards has them.

TWO TOOLS FIRST

`git` · `uv`

Both on the machine before you paste. The assistant checks, and stops if one is missing.

NOTHING TO CLONE

`init-researcher` · `init-strategy` ·
`init-example`

Three skills that each make a folder. Today you run the first two.

ONE CATCH

A skill is discoverable in a *new* session, never the one that installed it. Close it and reopen.

RUN THE INTERVIEW.

Fifteen minutes, in the room, together. It writes `RESEARCHER.md`, scaffolds every folder that is missing, and writes the agent file.

`init-researcher` Run it again with force to start over. You will want to, once you know what you actually wanted.

01**WHO IT WORKS FOR**

Your name, what you invest, and the domains it should follow — coding, math, finance, macro.

02**HOW IT SPEAKS**

Terse or thorough. And where it may push back: on evidence, never on taste.

03**WHAT YOU READ FOR**

Four or five open questions — and, for each, what would change your mind. Every source is read against them, by number.

04**THE NON-NEGOTIABLES**

What it is never allowed to do. The answer that makes it useful instead of agreeable.

THE NON-NEGOTIABLES.

Four lines from a real RESEARCHER.md. Write your own version of these before you write anything else.

- 01 Every number about a strategy comes from the Lab's libraries. The researcher never computes one itself.
- 02 A hypothesis is written before its test, and every prediction in it cites a source. No source, no prediction.
- 03 A strategy graduates only against criteria written down beforehand. Never on a good month.
- 04 The design of an experiment is challenged before it runs, not after it fails. Find the flaw while it is still cheap.

HOW IT CITES A claim with no source is written as a lead, never as a fact.

FEED IT SESSION 02.

The ten works on the two timelines, in order — each answers the one before it. Drop the PDFs into Sources/Papers/ and read them one at a time.

read plan first · your go · then it writes · contradictions flagged, never overwritten

TIMELINE 01 · THE ARGUMENT

1970

Fama · Efficient Capital Markets: A Review of Theory and Empirical Work

1979

Kahneman & Tversky · Prospect Theory

1980

Grossman & Stiglitz · On the Impossibility of Informationally Efficient Markets

1997

Shleifer & Vishny · The Limits of Arbitrage

2004

Lo · The Adaptive Markets Hypothesis

TIMELINE 02 · THE EVIDENCE

2011

Khandani & Lo · What Happened to the Quants in August 2007?

2016

Daniel & Moskowitz · Momentum Crashes

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McLean & Pontiff · Does Academic Research Destroy Stock Return Predictability?

2016

Harvey, Liu & Zhu · ... and the Cross-Section of Expected Returns

2023

Jensen, Kelly & Pedersen · Is There a Replication Crisis in Finance?

WHY THESE TEN Four of them are the reason the rest of this session is about discipline: crowding, crashes, post-publication decay and failed replication.

ONE PACKAGE, EVERY SKILL.

Every one that writes shows its plan first and waits for your go. `apm update -g` refreshes them all.

MAKE A FOLDER

`init-researcher`

The interview. Writes RESEARCHER.md and the agent, scaffolds the home.

`init-strategy`

A new strategy from the template: eight steps as folders, empty bibliography.

`init-example`

The same scaffold, filled in — a worked strategy to study. Homework.

`update [check]`

Brings a new version, keeps your RESEARCHER.md, Philosophy and Knowledge.

BUILD AND ASK

`read`

Sources into Knowledge, a chapter at a time — you pick which chapters. In a strategy, into its Bibliotheca.

`query <question>`

Answers from the library. Every claim cited, gaps named. Fires on its own.

`objective [strategy]`

Drafts the main idea and its claims — before any paper, then again from the notes.

`blueprint · brainstorm · challenge`

Thesis and predictions; the next thing to try; a finished experiment checked against its own blueprint.

KEEP IT HONEST

`audit [deep]`

Read-only: links, duplicates, stale index, orphans. `deep` adds contradictions.

`refine <path>`

A voice-preserving editor pass over your own writing. Diff first.

`refresh-index`

Rebuilds the index from what is actually on disk.

`teach <topic>`

A multi-session tutor, grounded in what you have read.

TODAY, IN THIS ORDER

`init-researcher` → `init-strategy` → `objective` → the universe → the Curator → `universe.ipynb` → stop

YOUR STRATEGY

init-strategy · the order of work · the prefix law

ONE COMMAND, ONE STRATEGY.

Open your assistant in the folder that will hold your strategies and run it. It asks for the name, then builds the template around it.

`init-strategy`

The KaxaNuk Strategy Template, copied from the package on your machine: the eight steps as folders, `OBJECTIVE.md` waiting for claims, `BIBLIOGRAPHY.md` and `LOG.md` empty. No code on *main*.

01

YOURS, AND PRIVATE

Not a fork. A fresh repository you own — push it to your own GitHub, private. One folder per strategy, as many as you like.

02

INVITE THE RESEARCHER

```
claude --add-dir <your researcher folder>
```

Its library comes with it; the skills are already there. Work lands in the strategy, nothing flows back home.

03

THE EXAMPLE IS THE SAME SHAPE

`init-example` builds a worked strategy beside yours. You will run it this week and read it for signals — not copy it.

THEN, IN A NEW SESSION Open the assistant in the strategy folder with the researcher added, and the first command it accepts is `objective`.

OBJECTIVE FIRST.

Eight items, the template's order. A command asked for out of order names the item that comes first, and stops. Today is items 1. 3 and half of 4.

1	THE OBJECTIVE before any paper is read	objective — claims from your own words, each untested, with the question that settles it	OBJECTIVE.md	TODAY
2	FINE-TUNE THE OBJECTIVE reading for each claim	read, one note per paper naming the claim it serves; then objective again	Bibliotheca/	HOMEWORK
3	THE INVESTABLE UNIVERSE delisted names included	the cohort universe — the KN US Equity Benchmark, one list for everyone	Universe/...csv	TODAY
4	THE DATA curator · universe notebook · refinery · analyzer	Curator and universe.ipynb today; Refinery and Analyzer on the example this week	Data/	TODAY
5	BENCHMARK, THEN BLUEPRINT before the rule	name the benchmark this week; blueprint and portfolio in session 04	Experiments/Exp_1/	SESSION 04
6	BROAD READING, BRAINSTORM	for what the blueprint left open	BRAINSTORMING_N.md	LATER
7	THE CYCLE	portfolio · backtest · attribution	the experiment notebook	LATER
8	RESULTS	kept or rejected	RESULTS.md	LATER

THE PREFIX IS THE ADDRESS.

It tells you which stage owns a column — and therefore which file to open when it is wrong.

m_*	THE PROVIDER, VIA THE CURATOR Raw market data — what arrived.	Change it: never.
C_*	CURATOR CUSTOM CALCULATIONS One security's own history. Nothing else is visible.	Change it: you need a new per-security quantity.
r_*	REFINERY CUSTOM CALCULATIONS Securities against each other, per date.	Change it: you need a rank, a breadth reading, or a fitted model.
current_*	JOINED FROM THE SECURITY MASTER Today's classification — <i>not</i> point-in-time.	Change it: you group or report by something new. Never select on it.

THE TRAP Selecting on **current_*** is survivorship bias with a respectable-looking column name. It knows today's sector for a company that was something else in 2014.

FROM OBJECTIVE TO DATA

the Curator, live · stop at the security master · trust, then build

ONE INTERFACE, EVERY PROVIDER.

Open source, on PyPI, no account needed. It turns provider chaos into one homogenised time series per identifier.

CONFIGURED IN THE REPO

No spreadsheet. Identifiers come from `Investable_Universe.csv`, dates and columns from the strategy's own parameters — plain files, versioned with the strategy.

UNIFIED ACCESS

One interface across data providers and your own in-house files, with the tags homogenised.

PREBUILT CALCULATIONS

Returns, ratios, trend, volatility, volume — and outlier adjustments — ready to switch on.

YOUR OWN **C_*** FUNCTIONS

Custom calculations extend it. This is where your per-security feature inputs are written.

FOUR COMMANDS.

All from the strategy folder. Environment, key, configuration, run — then the notebook.

01

ENVIRONMENT

```
uv sync
```

Installs the Data Curator and the FMP extension the template declares. One command, one lock file.

The Python version is the first thing that bites. 3.13.

02

THE KEY

```
cp Config/.env.template  
Config/.env
```

Open it and paste your Financial Modeling Prep key. Nothing else goes in there.

Never print a value from .env — an exposed key is rotated, not edited out.

03

THE CONFIGURATION

```
Universe/Investable_Universe.csv · Data/Curator/
```

The universe CSV supplies the identifiers; the date range, provider `fmp` and the output columns live in the strategy's own parameters. Ask the assistant to write them from your objective — as in the `liquid-golden-cross` example.

Start with the cohort universe and ten years. Widen later.

04

THE RUN

```
kaxanuk.data_curator run
```

One structured file per identifier, aligned on one calendar, in `Data/Curator/Time_Series/`.

Nothing under `Data/` is committed. A regenerable file is not a backed-up file.

THEN, FIFTH `universe.ipynb` — profiles what the Curator downloaded and writes the security master the Refinery will join. Run it, look at the coverage, and stop.

STOP HERE, ON PURPOSE.

ON DISK WHEN YOU LEAVE

- A researcher with a name and a `RESEARCHER.md` you wrote
- A strategy folder, pushed to your own private repository
- `OBJECTIVE.md` — a first pass of claims, each marked untested
- `Universe/Investable_Universe.csv` — the cohort universe
- `Data/Curator/Time_Series/` — one file per identifier, ten years
- The security master, written by `universe.ipynb`

Before you leave, every item on the left exists on your machine. Every item on the right is deliberately not done yet.

NOT TODAY

- The Refinery — you will run it on the example this week
- The Analyzer — the gate every signal passes before session 04
- The benchmark choice — homework, in words, in `BRAINSTORMING_1.md`
- The blueprint, the portfolio, the backtest — session 04

A pipeline you ran once, end to end, on data you understand beats a portfolio you cannot explain. That is why we stop.

TRUST, THEN BUILD.

Six checks on what the Curator wrote, before any feature is built on it. Each one is a rule applied to every security and every date — never a judgement call on one row.

01

SPLITS AND DIVIDENDS

Adjusted prices, or one unadjusted split invents a –50% day and a momentum signal that never existed.

02

THE DEAD ARE PRESENT

Delisted names in the universe with their last price. A universe of survivors backtests beautifully and trades badly.

03

THE AVAILABILITY DATE

Fundamentals carry three dates. The one you may use is when the market could have read it — not the period end.

04

OUTLIERS, BY RULE

Statistical: is it unusual (z-score, MAD)?
Financial: is it possible (negative price, volume of zero for a year)? Both, then a rule.

05

GAPS AND STALE PRICES

A price that does not move for twenty days is not a calm stock, it is a missing feed. Coverage first, then signal.

06

EVERY CORRECTION LOGGED

Whatever you change lands in `Data_Issues.csv` with the rule that changed it. A cleaning step you cannot explain is a result you cannot defend.

WHAT TO LOOK FOR IN THE EXAMPLE

the menu of signals · the six ways a number lies to you

PICK ONE, NOT EIGHT.

Your objective already implies one of these. Build that one properly before you build a second.

ALPHA SIGNALS

Earnings surprise — abnormal returns following an unexpected result, and the drift after it.

Short-term reversal — mean reversion after an extreme short-horizon move.

Residual momentum — trend persistence that survives removing factor exposure.

Quality — profitability and balance-sheet strength, priced slowly.

Accruals — mispricing tied to low earnings quality.

Sentiment — predictive content in news, filings and textual tone.

Event-driven — temporary mispricing around M&A, buybacks and guidance.

RISK FACTORS

Market — exposure to the asset class moving at all.

Beta — sensitivity of the security to that benchmark.

Sector — exposure to industry-specific economics.

Size — market capitalisation.

Value — price relative to fundamentals.

Momentum — the persistent trend, as a premium rather than an edge.

Volatility — exposure to return variability.

Growth — exposure to strong fundamental expansion.

NOTICE Momentum and quality appear on both sides. Which one yours is depends entirely on whether attribution can explain it away.

SIX WAYS IT IS A LIE.

When a result surprises you on the upside, one of these is usually why. Check them in this order.

LOOK-AHEAD

A value indexed on a date before you could have had it. The commonest, and the most flattering.

SURVIVORSHIP

Your universe is who is still listed. The delisted names are the ones the strategy would have bought.

SELECTION ON **CURRENT_***

Filtering history by today's classification. Survivorship with a respectable column name.

UNDECLARED SWEEPS

Forty lookbacks tried, one reported. The p-value you quote is for the one test you pretended to run.

UNCOSTED TURNOVER

A signal that rebalances daily on illiquid names is a spread-payment schedule, not a strategy.

BORROWED ALPHA

It works, and it is sector beta. Real, tradeable, and not worth a fee. Attribution settles this in session 05.

ASK YOUR RESEARCHER /query which of these six does my feature risk, and what did we read about it?

UP NEXT

NEXT: PORTFOLIO CONSTRUCTION.

Session 04 turns your signals into historical portfolios that have to beat the benchmark you name this week. It starts from your `OBJECTIVE.md`.

TO DO.

Four deliverables, all committed to your strategy repository.
Session 04 cannot start without the last two.

01 RUN THE EXAMPLE

Study it for ideas, not for answers.

- `init-example`, beside your strategy
- Curator → `universe.ipynb` → Refinery → Analyzer, end to end
- Read its `OBJECTIVE.md` and blueprint: which signals, and why
- Write down the two you would steal, and the one you would not

02 YOUR CLAIMS

The objective, fine-tuned by reading.

- `objective` — a first pass from your own words
- read at least two of the ten works into your researcher
- `objective` again — the evidence rewritten from the notes
- Each claim numbered, present tense, with what would kill it

03 YOUR BENCHMARK

What "beat" means, in one sentence.

- An index, an ETF or an equal-weight book of the universe
- Why that one, given your objective
- The first entry of `BRAINSTORMING_1.md` — Experiment 1 *is* the benchmark

04 TWO OR THREE SIGNALS

Candidates, not ten. The ones your claims imply.

- Each one named, and the claim it serves by number
- Each one citing the note it came from
- Its availability lag stated, in words
- One sentence on which of the six lies it is exposed to

WHERE IT GOES Claims, benchmark and signals into your strategy repository. The researcher stays on your own machine — your library is private to you.

THE READING LIST.

Read two. Compile them. If you only read one, make it Arnott, Harvey & Markowitz — it is a checklist disguised as a paper.

THE DISCIPLINE · WHY A FEATURE MUST BE DECLARED BEFORE IT IS TESTED

ARNOTT, HARVEY & MARKOWITZ

2019 A Backtesting Protocol in the Era of Machine Learning

HARVEY, LIU & ZHU

2016 ... and the Cross-Section of Expected Returns

MCLEAN & PONTIFF

2016 Does Academic Research Destroy Stock Return Predictability?

THE CRAFT · HOW THE COLUMNS ARE ACTUALLY BUILT

PALEOLOGO

2025 The Elements of Quantitative Investing
2021 Advanced Portfolio Management

GRINOLD & KAHN · DIXON

1995 Active Portfolio Management
2020 Machine Learning in Finance: From Theory to Practice

VIAFORE

2021 Robust Python
For the custom calculations you are about to write and then rely on for two years.

TWO FOLDERS. ONE PACKAGE.

What you know, and what you are testing. A feature that cannot name its source is not finished.

```
github.com/KaxaNuk/KaxaNuk-Researcher
```

Paste the URL, name it. One per person.

```
init-strategy · init-example
```

From the same package. One per strategy.

QUESTIONS research@kaxanuk.mx

NEXT

SESSION 04

TOPIC

**PORTFOLIO
CONSTRUCTION**

YOU LEAVE WITH

**HISTORICAL
PORTFOLIOS FROM
YOUR SIGNALS,
MEASURED AGAINST
YOUR BENCHMARK.**

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